Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Git and Github Masterclass

Course Code: C171

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 3.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	3 July 2026	Previous course schedule.	Dr Alfred Ang
2.0	15 August 2026	Rebuilt two-day plan aligned to the January 2026 GH-900 domains, current slides, eight labs and WA plus Case Study assessment.	Dr Alfred Ang
2.1	15 August 2026	Expanded to fourteen learn-by-doing labs with dedicated Git command, GitHub Desktop, conflict, recovery and exam-readiness practice.	Dr Alfred Ang
2.2	18 August 2026	Expanded Git deep-dive, integrated legacy master-trainer coverage across all seven GH-900 domains, added Git command cheat sheet to deck and Learner Guide.	Dr Alfred Ang
3.0	4 October 2026	One-day edition for Git and Github Masterclass (C171): all seven domains and fourteen labs retimed to 9:30 am – 5:30 pm.	Dr Alfred Ang

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Course Information

Course Title	Git and Github Masterclass
Course Code	C171
Duration	1 day · 7.5 instructional hours
Daily Timing	9:30 am – 5:30 pm (30-minute lunch break)
Mode	Instructor-led with fourteen GitHub labs and Excel evidence workbooks
Trainer	Dr Alfred Ang

Learning Outcomes

- LO1: Analyze GitHub components and coordinate release scheduling with collaborators to align processes.
- LO2: Select appropriate Git scripts for integrating and deploying software products.
- LO3: Configure software products and deploy releases using Git configuration tests.
- LO4: Diagnose issues identified during GitHub configuration testing by interpreting configuration test results.
- LO5: Identify potential improvements to software configuration, deployment processes, and code elements.
- LO6: Implement modifications to software products and processes for improved functionality.

Course Schedule

Day 1 · 9:30am – 5:30pm — Git Foundations, Collaboration, Automation and Governance

Time	Minutes	Type	Topic / Activity	Slide reference
9:30–9:45	15	Admin	Welcome, trainer profile, ground rules and course portal	2–8
9:45–10:10	25	Concept	GH-900 blueprint, Git areas, file states and toolchain	—
10:10–10:35	25	Lab	Lab 01 — Install Git, Git Bash and GitHub Desktop	27
10:35–11:00	25	Lab	Lab 02 — Create and	29

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			Clone a Training Repository	
11:00–11:30	30	Lab	Lab 03 — Status, Diff, Add, Commit and Log	55
11:30–11:45	15	Concept	Domain 2 — branch pointers, merge strategies and remotes	63
11:45–12:15	30	Lab	Lab 04 — Create Branches and Merge Changes	57
12:15–12:40	25	Lab	Lab 05 — Synchronise with Fetch, Pull and Push	59
12:40–13:10	30	Lunch	Lunch break	—
13:10–13:35	25	Lab	Lab 06 — Resolve a Merge Conflict and Recover Safely	61
13:35–13:50	15	Concept	Domain 3 — pull requests, issues and collaborative change	78
13:50–14:15	25	Lab	Lab 07 — Open a Pull Request and Complete Code Review	74
14:15–14:40	25	Lab	Lab 08 — Coordinate Work with Issues, Discussions and Pages	76
14:40–14:55	15	Concept	Domain 4 — Actions, Copilot, Codespaces and github.dev	94
14:55–15:20	25	Lab	Lab 09 — Build	90

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			and Diagnose a GitHub Actions Check	
15:20–15:45	25	Lab	Lab 10 — Compare Codespaces, github.dev and Copilot	92
15:45–16:00	15	Concept	Domain 5 — GitHub Projects and release coordination	105
16:00–16:20	20	Lab	Lab 11 — Plan and Track a Release with GitHub Projects	103
16:20–16:35	15	Concept	Domain 6 — privacy, security and administration	118
16:35–16:55	20	Lab	Lab 12 — Secure and Administer a Repository	116
16:55–17:10	15	Lab	Lab 13 — Contribute through a Fork and Explore the Community	127
17:10–17:25	15	Lab	Lab 14 — Build a GH-900 Readiness Plan	129
17:25–17:30	5	Recap	Course recap, retrieval practice, Q&A and course feedback	137–144

Total instructional time: 450 minutes (7.5 instructional hours, 9:30am – 5:30pm), with a single 30-minute lunch break.

Domain and Lab Alignment

Exam domain	Weighting	Aligned labs	First slide
Understand Git and GitHub Basics	25–30%	Lab 01, Lab 02	10
Work with GitHub	10–15%	Lab 03, Lab 04,	32

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Repositories		Lab 05, Lab 06	
Collaborate Using GitHub	10–15%	Lab 07, Lab 08	64
Apply Modern Development Practices	10–15%	Lab 09, Lab 10	79
Manage Projects with GitHub	5–10%	Lab 11	95
Understand Privacy, Security and Administration	10–15%	Lab 12	106
Explore the GitHub Community	5–10%	Lab 13, Lab 14	119

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